

Sherif Gerges

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Summary

- Human geneticist working at the intersection of statistical genetics, single-cell genomics, and protein structure. My research integrates large-scale genetics with structural modeling to uncover mechanisms driving neuropsychiatric disorders, with a focus on using genetics to discover novel biology for drug discovery.

Education

PhD	Harvard Medical School , Genetics and Genomics • Advisors: Mark Daly and Steve McCarroll	2017 – 2025
BS	Montclair State University , Molecular Biology; minors in Computer Science and Mathematics • Magna Cum Laude	2010 – 2013
	German University in Cairo, Egypt , Pharmaceutical Sciences (transferred)	2008 – 2010

Experience

Broad Institute of MIT and Harvard , Postdoctoral Researcher • Labs of Mark Daly and Steve McCarroll • Developing structure-based statistical methods for interpreting rare missense variation in neuropsychiatric disease, validated with cryo-EM	Cambridge, MA 2025 – present 1 year
Harvard Medical School , PhD Student • Thesis advisors: Mark Daly and Steve McCarroll • Developed a multi-omic strategy integrating large-scale single-cell transcriptomics, proteomics, and human genetics to uncover genes and pathways implicated in neuropsychiatric disorders, with emphasis on schizophrenia and autism spectrum disorders • Developed a statistical method that leverages AlphaFold3-predicted protein structures to reveal non-random mutation clustering and infer disease mechanisms at the domain level, demonstrating a powerful predictor of missense variant effects and an orthogonal tool for variant prioritization (manuscript in preparation) • Initiated and led a multi-institutional collaboration co-designing CRISPR-based cellular models and protein-based biochemical assays to test missense variants from exome studies of schizophrenia and ASD, coordinating the labs of Mark Daly, Steve McCarroll, and Hilary Finucane (Broad Institute), Poul Nissen (Aarhus University), and Jonathan Weissman (MIT) • Collaborated on a high-throughput droplet-based barcoding strategy recording RNA co-occurrence across thousands of individual synapses, revealing transcriptomic compartments that organize synaptic biology and are enriched for neuropsychiatric genetic signal (manuscript in preparation) • Collaborated on integrative genetic analyses demonstrating that variation in cis-regulatory sequences of granulocytes and monocytes accounts for a substantial proportion of tuberculosis heritability • Collaborated with the Autism Sequencing Consortium on single-cell transcriptomic expression patterns of newly identified ASD genes and variant prioritization	Boston, MA 2017 – 2025 8 years

Acurasset Therapeutics , Co-founder - Co-founded with Professor Alexander Ploss, using advanced HBV infection models to identify small-molecule inhibitors of a previously unexploited step in the viral lifecycle - Hired a cross-functional team, including a contract research organization and a medicinal chemist from Harvard's Therapeutics Initiative, to advance the chemical development program - Lead all fundraising and operations; raised over \$500,000 in venture capital and non-dilutive grants	2022 – present 4 years
Third Rock Ventures , Computational Consultant - Lead computational consultant for the founding team of an early-stage venture in immuno-oncology and neuroimmune disease (now Asher Bio) - Built a statistical pipeline processing large transcriptomic datasets to identify highly specific single-cell RNA-seq cell-surface markers across indications - Work contributed to Asher's lead program, AB248, an engineered IL-2 immunotherapy selectively targeting CD8+ effector T cells	2020 – 2021 1 year
Princeton University , Research Assistant, lab of Alexander Ploss Molecular cloning, cell culture, and viral production to engineer hepatitis B, C, and D virus strains capable of infecting humanized mouse models	2014 – 2016 2 years
California Institute of Technology , Research Assistant, lab of Kai Zinn - Co-developed a stochastic block model simulating the evolution of modular biological networks, applied to yeast proteomics, mouse brain connectivity, and the <i>C. elegans</i> connectome - Modeled a 94-protein fat-regulation network in yeast, showing small-world (Watts-Strogatz) topology and modular organization supported by genetic and pharmacological validation	2012 – 2014 2 years

Selected Talks

- American Society of Human Genetics (ASHG), 2019-2021** - Human single-cell transcriptomes identify cell types and states relevant to brain-related complex traits
- World Congress of Psychiatric Genetics, 2021** (invited talk; oral talk finalist) - An atlas of human cell-type transcriptomes and their roles in schizophrenia
- 6th Neurological Disorders Summit, 2020** (plenary talk) - Human single-cell transcriptomes identify cell types and states relevant to brain-related complex traits

Honors & Awards

- 2023:** Princeton University IP Accelerator Award
- 2022:** Nucleate-Petri VC Startup Winner (\$200,000 venture funding award)
- 2022:** John Maraganore Scientific Entrepreneurship Award (semi-finalist)
- 2022:** Harvard Innovation Lab Award
- 2021:** BioNJ BioPartnering Company Presentation Award